

Wang YuHeng

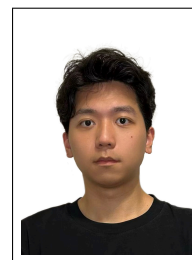
Data Science Undergraduate

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Professional Summary

Final-year Data Science undergraduate at HKUST with a strong foundation in **machine learning, computer vision, statistical modeling, and programming**. Trained a U-Net model for cardiac detection and segmentation on X-ray images, achieving 90%+ accuracy and a 0.7992 F1-score. Conducted research with Prof. Gary Chan on floorplan-guided 3D reconstruction using 2D floorplans as structural priors. During a tech internship, collected 3D food point-cloud data and built a volume-estimation algorithm from scratch, reducing MAPE from 25.4% to 12.36%. Also self-studied full-stack development with working knowledge of both front-end and back-end engineering. Currently seeking opportunities in data analytics and AI/CV research.

Education

09/2023 – 06/2027 (expected) **B.Sc. in Data Science**, *The Hong Kong University of Science and Technology (HKUST)*, Hong Kong

GPA 3.733 / 4.3 (Major GPA: 3.834)

Ranking Top 10%

Relevant Coursework

- **Mathematics & Statistics**: Honours Calculus I/II [A], Multivariable Calculus [A+], Statistics [A], Honours Probability [A+], Mathematical Analysis [A], Matrix Computation [A], Matrix Differentiation [A], Bayesian Statistics [A], Statistical Inference [A]
- **Computer Science & ML**: C++ & Data Structures [A], Discrete Mathematics [A+], Machine Learning [A], Algorithms [B+]

Research Experience

Sep 2025 – Dec 2025 **Undergraduate Researcher – 3D Point Cloud Reconstruction**, *HKUST UROP / Course Project*, Hong Kong

Developed a floorplan-guided framework that aligns and fuses multi-view point clouds using 2D floorplans as structural priors, improving geometric consistency and scale normalization for indoor 3D scenes (digital twins, robotics, BIM).

Projects & Internship Experience

2026 **Alpha Factor Mining Pipeline – WorldQuant BRAIN**

Built an end-to-end pipeline that automated simulation, result polling and submission planning on the BRAIN platform (**1,354 simulations; 15 alphas submitted, all live; best fitness 3.88**).

Reverse-engineered **six** of the platform's admission rules from my own results, so candidates could be judged *before* submission; showed the co-existence constraint is a **maximum-clique problem** – 22 alphas cleared the gates but only **4** could be submitted together.

2026 **Kaggle Competition – Kaggriculture: Two-Player Farming-Economy Agent**

Built a 720-turn two-player farming agent for the Kaggle Kaggriculture ladder, reaching a ladder Elo of **2452.2** (about 850th of ~8,000 teams).

Built a paired multi-seed tournament harness (dual-seat, ≥ 500 games per arm, two independent seed blocks) and used it to isolate one decisive constant, lifting the win rate against the strongest local opponent from 37.5% to **93.1%**.

Summer 2025 **Research Intern – Medical Image Segmentation**, *Seoul National University (SNU–HKUST Sponsored Project)*, Seoul, Korea

Built and trained a U-Net model for automated heart detection, localization, and segmentation on chest X-ray images, achieving 90%+ accuracy and a 0.7992 F1-score.

2026 – **FYP – Towards a Personal Embodied AI Agent with Reinforcement Learning**
Present

Developing a perception–action pipeline in PyTorch with reinforcement learning, enabling an embodied agent to perform visual search and manipulation tasks from natural-language instructions.

Jul – Aug 2026 **Computer Vision Intern – Food Volume Estimation**, *AlxHealthcare Lab (Shenzhen Zhijie Exploration Technology Co., Ltd.)*, Shenzhen, China

Built a food-volume estimation algorithm from scratch for a nutrition-estimation application; reduced MAPE to **12.36%** on the test set (vs. $\sim 20\%$ with the company's original method) after one and a half months of development.

Explored state-of-the-art 3D reconstruction models (VGGT-Omega, Pi3X, DA3, MAST3R, etc.) and gained hands-on experience with the full algorithm development cycle.

Aug 2026 **Data Visualization Project**

Created a visualization exploring AI's impact on the job market (<https://github.com/TheLogarhythm/Career-a-vis-AI>).

Gained hands-on experience with data visualization tools including Datawrapper, Tableau, pandas, iNZight, and RAWGraphs.

Ongoing **Full-Stack Development (Self-Study) – Personal Website**

Self-studied full-stack development and built a personal website to present my experience and showcase projects (in progress).

<http://120.79.180.47>

Certificates

WorldQuant Challenge – Bronze, Silver and Gold Level (WorldQuant).

Skills

Programming Languages Python (most proficient), C++, Java, R

Languages Mandarin (native), English

Front-End React, Vite, npm